Note

If we change exactly one of the four given lengths, then we can make these triangles similar.

Example 4.3 Observe Fig.4.18 and find $\angle P$.

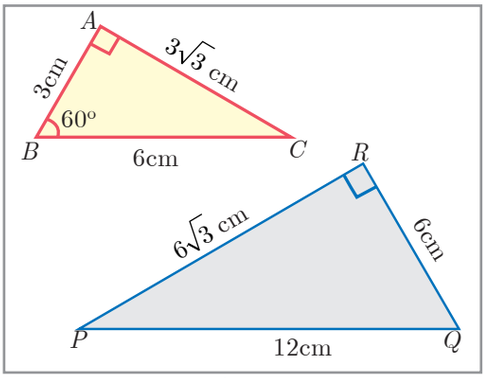


Fig. 4.18

Solution In ΔBAC and ΔPRQ, $\frac{AB}{RQ}=\frac{3}{6}=\frac{1}{2}$;

$$\frac{BC}{QP}=\frac{6}{12}=\frac{1}{2};\ \frac{CA}{PR}=\frac{3\sqrt{3}}{6\sqrt{3}}=\frac{1}{2}$$

Therefore, $\frac{AB}{RQ}=\frac{BC}{QP}=\frac{CA}{PR}$

By SSS similarity, we have $\Delta BAC\sim\Delta QRP$

$\angle P=\angle C$ (since the corresponding parts of similar triangle)

$\angle P=\angle C=180°-(\angle A+\angle B)=180°-(90°+60°)$

$\angle P=180°-150°=30°$

Example 4.4 A boy of height 90cm is walking away from the base of a lamp post at a speed of 1.2m/sec. If the lamppost is 3.6m above the ground, find the length of his shadow cast after 4 seconds.

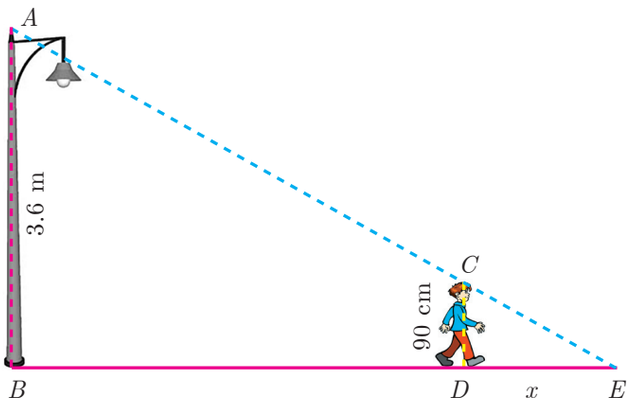


Fig. 4.19

Solution Given, speed = 1.2 m/s,

time = 4 seconds

distance = speed × time

= 1.2×4 = 4.8 m

Let x be the length of the shadow after 4 seconds

Since, $\Delta ABE\sim\Delta CDE$, $\frac{BE}{DE}=\frac{AB}{CD}$ gives $\frac{4.8+x}{x}=\frac{3.6}{0.9}=4$ (since 90 cm = 0.9 m)

$4.8+x=4x$ gives $3x=4.8$ so, $x=1.6$ m

The length of his shadow $DE=1.6$ m

Example 4.5 In Fig.4.20 $\angle A=\angle CED$ prove that $\Delta CAB\sim\Delta CED$. Also find the value of x.

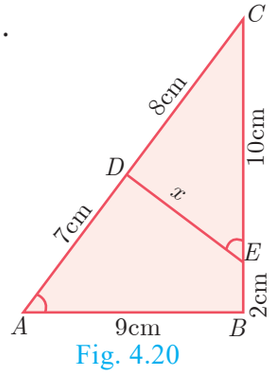


Fig. 4.20

Solution In ΔCAB and ΔCED, $\angle C$ is common, $\angle A=\angle CED$

Therefore, $\Delta CAB\sim\Delta CED$ (By AA similarity)

Hence, $\frac{CA}{CE}=\frac{AB}{DE}=\frac{CB}{CD}$

$\frac{AB}{DE}=\frac{CB}{CD}$ gives $\frac{9}{x}=\frac{10+2}{8}$ so, $x=\frac{8\times9}{12}=6$ cm.

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